

Urinary excretion of total isothiocyanates from cruciferous vegetables shows high dose-response relationship and may be a useful biomarker for isot...

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BACKGROUND: Isothiocyanates (ITCs), hydrolysis products from glucosinolates, are a family of biologically active compounds originating from cruciferous vegetables. Many ITCs are assumed to have cancer preventive effects and to further evaluate these potential health effects, reliable biomarkers of ITC exposure are needed. AIM OF THE STUDY: In this study we investigated the ability of urinary ITC excretion to reflect a low or high daily intake of cruciferous vegetables. METHODS: The design was a controlled human crossover study (n = 6). Subjects consumed a self-restricted glucosinolate-free diet 48 h before the study-day where a basic diet supplemented with 80 or 350 g of mixed cruciferous vegetables was consumed. All urine was collected in intervals during the 48 h period after ingestion of the cruciferous vegetables. Total ITC in the cruciferous mixture and total ITC and their metabolites in urine was quantified as the cyclocondensation product of 1,2-benzenedithiol by high performance liquid chromatography. RESULTS: The total urinary excretion of ITCs correlated significantly with the two doses of ITC from diets with high or low cruciferous content ($r(s) = 0.90$, $P < 0.01$). The fraction of urinary ITC excreted was $69.02 \pm 11.57\%$ and $74.53 \pm 8.39\%$ of the amounts ingested for 80 and 350 g cruciferous vegetables, respectively. CONCLUSION: The results in this study indicate that the urinary excretion of ITCs, measured by use of the cyclocondensation reaction, is a useful and precise tool that may be used as a biomarker of ITC exposure in population based studies.